

Design Specification — VBF-T6R2-DS-001

Design Specification — 4C-Capable Smartphone Cell (VBF-T6R2-DS-001)

Case: exp/t6_r2 · **Status:** criteria achieved (DFN-verified) · **Date:** 2026-08-26

1. Objective

Design a battery for a smartphone: volumetric energy density ≥ 950 Wh/L, 4C fast charge without lithium plating, maximum temperature ≤ 50 °C, anode SEI thickness ≤ 500 nm after 100 cycles, voltage plateau ≥ 4.1 V.

2. Final design (candidate R12a_h140, params `cell/r12\_a\_h140.json`)

Parameter	Value	Role
Electrode system	LNMO high-voltage spinel (4.7 V class) / gra	Plateau ≥ 4.1 V (NMC811 mid
Negative electrode thickness	120 μm	N/P margin $\rightarrow$ plating headro
Negative particle radius	1.5 μm	Less anode diffusion overpot
Positive particle radius	3.0 μm	Lower cathode diffusion over
EC diffusivity	5.0e-19 m ² /s	Dense FEC/VC-class SEI coati
SEI kinetic rate constant	3.0e-13 m/s	SEI growth suppression
Cell cooling surface area	0.0106 m ²	Double-sided thin-pouch exp
Total heat transfer coefficient	140 W/m ² ·K	Vapor-chamber-grade therm

All other parameters: Chen2020 baseline + LNMO system file overrides.

3. Criteria compliance (precise DFN model, mechanical evaluation — log round 12)

Criterion	Threshold	Result (DFN)
Volumetric ED	≥ 950 Wh/L	1096.0 Wh/L
Voltage plateau (midpoint)	≥ 4.1 V	4.1301 V
Max temperature (4C, 45 °C amb.)	≤ 50 °C (323.15 K)	322.70 K
Lithium plating (4C)	none (min anode potential > 0)	+0.0382 V, plated=False
SEI after 100 cycles	≤ 500 nm	415.6 nm

SPMe cross-check (cell/r12_a_energy_spme.json, cell/r12_a_4c45_spme.json): ED 1106.3 Wh/L, midV 4.2202 V, T 320.30 K, plated False — consistent direction, DFN is the gate.

4. Key design decisions (traceable to log rounds)

- **R1→R2 ceiling escalation**: Chen2020 NMC811 midpoint plateau ~ 3.78 V (DFN) is structurally below 4.1 V $\rightarrow$ system switch to LNMO 4.7 V-class spinel (funnel entry).
- **R5**: EC diffusivity identified as the wired SEI coating lever by pybamm source inspection ("SEI solvent diffusivity" is a no-op; D_ec 2e-18 $\rightarrow$ 5e-19 $\square$ SEI 708 $\rightarrow$ 415 nm).
- **R6 (SPMe) / R8 (DFN)**: plating root cause = terminal anode-surface saturation cliff at the 4.7 V cutoff of the 4C charge from deep discharge; fixed by N/P margin (neg 120 μm) + small particles. Thinner positive and t $\square$ =0.4 each regressed plating (attributed, dropped).
- **R9–R12 (DFN thermal)**: DFN resolves internal gradients $\rightarrow$ heat 3.3 $\times$ SPMe. Cooling scaled via h + enlarged cooling area (0.0106 m²) after heat-reduction levers (kinetics) were exhausted; reaction-rate constant and separator thinning were tested and rejected/no-op.
- **R12**: final candidate passes all five criteria on DFN (verdict=pass, 5/5 checked).

5. Inert / rejected levers (documented)

- "SEI solvent diffusivity" — not wired in the ec-reaction-limited aging model (R4 c1).
- "Positive electrode reaction rate constant" — model uses exchange-current-density functions; scalar override is a no-op (R10a bit-identical output).
- Electrolyte diffusivity 5e-10 — deeper charge before cutoff $\rightarrow$ more heat (R8s2, rejected).
- Positive thickness 70 μm — plating regression on DFN (R8s3, rejected).
- Negative porosity 0.30 — thermal regression (R9s3, rejected).
- Constant electrolyte conductivity — froze Nyman2008 T-dependence $\rightarrow$ 359.7 K (R3 v4, rejected).

6. Hard constraints & caveats (honest)

- **Thermal interface is binding**: h ≥ 140 W/m²·K with ≥ 0.0106 m² exposed surface is a hard requirement (vapor chamber / chassis-coupled cooling). T margin is 0.45 K; passive cooling cannot meet T ≤ 50 °C at 45 °C ambient during 4C charge.
- **4C current basis**: the protocol current derives from the LNMO system file's nominal capacity (4.5 Ah); actual simulated capacity is 6.374 Ah $\square$ effective ~ 2.8 C. Recommend follow-up validation at true 25.5 A (4C of 6.374 Ah).
- **LNMO file artifact**: deprecated "Positive electrode diffusivity" override is ignored by pybamm in favor of "Positive particle diffusivity"; the intended cathode diffusivity is not applied. Aging per-cycle capacities are negative artifacts of the LNMO parameterization; SEI thickness is the reliable metric.

- ****True compute****: not run (real_compute=false pre-registered in entry 0); endorsement rests on the precise DFN model, not paper-grade DFT/MD (see endorse entry).